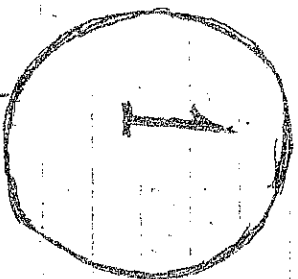


ANALOGUE ELECTRONICS IV

EEE 434

LECTURE NOTES ON
(Principles, characteristics and application of feedback amplifiers)

2005



EE3

min

au

Pre-lectures Interactive / self evaluation.

Explain the following terminologies:

- feedback in amplifiers (athletic/manual)
 - frequency response of an amplifier
 - Bandwidth of an amplifier
 - decibel
 - distortion in amplifiers (non linear, frequency and phase delay)
 - Interactive discussion will carry 20% of the total Exam
 - Applied knowledge will carry 10%
 - while lecture notes will carry 70%
- Total 100%

The Course comprises 4 topics as below:

1. Feedback amplifiers with emphasis on negative type feedback

2. Oscillators

3. Multivibrators/Pulse oscillators

4. Regulated power supplies

THE CONCEPT OF FEEDBACK IN AMPLIFIERS

Definition: Feedback is a process whereby the output of an amplifier is sampled and fed back to the input for necessary correction action. Feedback could either be negative or positive (degenerative or regenerative). In negative feedback the sampled output is now subtracted from the input before being amplified while reverse is the case in positive feedback. Positive feedback tends to make the amplifier unstable and hence breaks into oscillation which is useless as an amplifier. But this phenomenon is being harnessed as useful in oscillators.

Block diagram of a basic feedback amplifier (single loop)

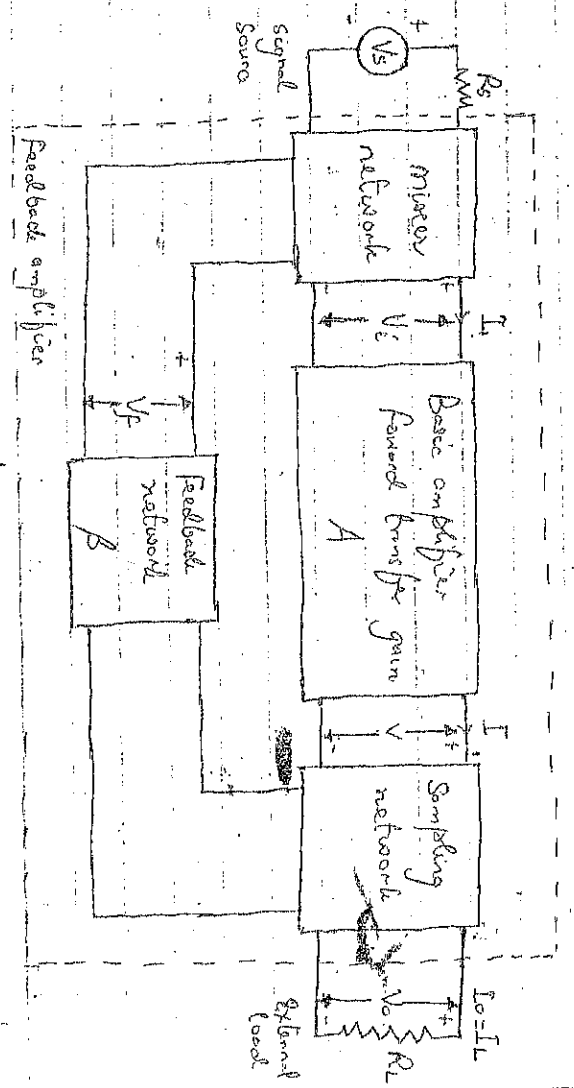


Fig. 1

The general diagram shows that the output voltage or current by means of a suitable sampling network and apply this signal to the input through a feedback two-port network. At the input the feedback signal is combined with external (source) signal through a mixer network and is fed into the amplifier proper. Also from above it can be seen that for negative feedback,

$V_i = V_s - V_f$ = difference signal

$V_f = V_o \beta$

and $V_o = AV_i$ ③
 Now from 1 and 2 and substituting in ③ yields

$$V_o = A(V_s - V_f)$$

$$V_o = A(V_s - V_o\beta)$$

$$V_o = AV_s - AV_o\beta$$

$$V_o + AV_o\beta = AV_s$$

$$V_o(1 + A\beta) = AV_s$$

$$A_{vf} = \frac{V_o}{V_s} = \frac{A}{1 + A\beta}$$

$$A_{vf} = \frac{A}{1 + A\beta}$$

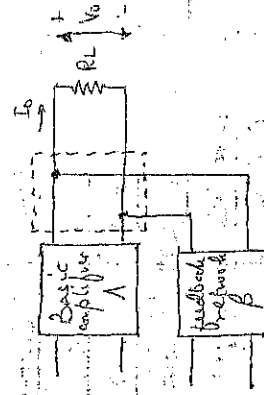
where A is the open loop gain
 and β is the feedback factor.

* Show that $A_{vf} = \frac{A}{1 - A\beta}$ for positive feedback.

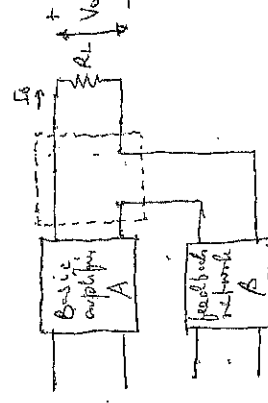
Note that in fig 1, the feedback network is really a passive two-port network which may contain resistors, capacitors and inductors. Very often it is simply a resistive configuration.

NEGATIVE FEEDBACK TYPES

Four sampling blocks are shown in fig 2, called shunt and series



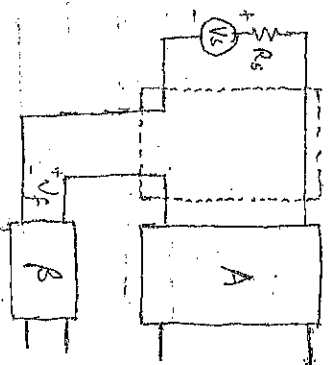
Voltage or Shunt



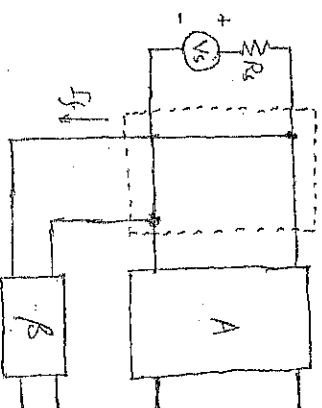
Current or Series

fig 2.

Several mixing blocks are shown in fig 3, called series and shunt feedback.



Series feedback



Shunt feedback

fig 3

The above will combine to give the following configurations

Such as:

- a) Series-Current feedback
- b) Series-Voltage feedback
- c) Parallel (shunt)-Current feedback
- d) Parallel (shunt)-Voltage feedback

* Ask the students to draw all the configurations.

EFFECTS OF APPLYING NEGATIVE FEEDBACK TO AN AMPLIFIER

ON -

✓ a) GAIN

The gain of an amplifier with feedback is reduced as compared with that of an amplifier without feedback. But this has been compensated for by the following:

b) GAIN STABILITY

The gain of the amplifier will vary due to aging, temperature replacement etc. To overcome this effect the amplifier must

✓ Prove

③

have feedback and can be shown from this formula that

$$A_{vf} = \frac{A}{1 + \beta A}$$

if the loop gain βA is made large enough by arrangement, $\beta A \gg 1$ and so

$$A_{vf} = \frac{A}{\beta A} = \frac{1}{\beta}$$

implying that the gain can be made to depend entirely on the feedback network and not on the active component. If the feedback network contains only stable passive elements, the improvement in stability may indeed be pronounced.

a) DISTORTION

The distortion could either be frequency, nonlinear and delay distortion.

i) Frequency/delay distortion

From $A_{vf} = \frac{1}{\beta}$, if the feedback network does not contain reactive elements, the overall gain is not a function of frequency. Under these circumstances a substantial reduction in frequency and phase shift (delay) distortion is obtained.

ii) Nonlinear distortion amplitude

Suppose that a large signal is applied so that the operation of the device extends slightly beyond its range of linear operation and as a consequence the output signal is slightly distorted. Negative feedback is now introduced and the input signal is increased by the same amount by which the gain is reduced, so that the output signal amplitude remains the same.

d) NOISE

It can be shown that the noise introduced in an amplifier is multiplied by the factor $\frac{1}{1 + \beta A}$ if feedback is employed. If $|1 + \beta A| \gg 1$, this would seem to represent

a considerable reduction in output noise.

F) INPUT/OUTPUT IMPEDANCE

Negative feedback which samples the output voltage regardless of how the ^{output} signal is returned to the input tends to decrease the output resistance. That which samples the output current will tend to increase the output resistance.

If the feedback signal is returned to the input in series to oppose the applied voltage, regardless of whether it is obtained by sampling the output voltage or current, tends to increase the input resistance. That in which the output signal is fed back to the input is parallel tends to decrease the input resistance.

* Ask student to draw all the 8 possible connections

F) BANDWIDTH / ~~AMPLITUDE~~ GAIN - BANDWIDTH PRODUCT

The bandwidth is extended by the use of negative feedback. whereas the gain-bandwidth product is not affected with or without feedback.

* demonstrate physically.

Evaluation

- 1) Assignment ✓
- 2) Test
- 3) Problems

Numerical Examples

Q1) An amplifier has an open-loop gain of 35. What is its voltage gain with 10% feedback?

$$\text{From } A_{vf} = \frac{A}{1+A\beta}$$

$$A = 35$$

$$\beta = 0.1$$

$$\therefore A_{vf} = \frac{35}{1+35 \times 0.1} = \frac{35}{4.5} = 7.7$$

from approximate formula when $A\beta \gg 1$

$$A_{vf} = \frac{1}{\beta} = \frac{1}{0.1} = 10$$

TUTORIALS

1. When voltage feedback is applied to an amplifier of gain 100, the overall gain falls to 50. Calculate the fraction of the output voltage fed back. If this fraction is maintained, calculate the value of the amplifier gain required if the overall stage gain is to be 75.

2. An amplifier having a gain of 500 without feedback has negative feedback applied which reduces the overall gain to 100. Calculate the fraction of the output voltage fed back. If due to aging of components, the gain without feedback falls by 20%, calculate the percentage fall in gain with feedback.

3. An amplifier having a gain of 100 has 9% voltage negative feedback applied in series with the input signal. Calculate the overall stage gain with feedback. If a supply voltage variation causes the gain with feedback to fall by 10%, determine the percentage change in gain without feedback.

(7)

- 4) With negative feedback, an amplifier gives an output of 10V with an input of 0.5V. When the feedback is removed, it requires 0.25V input for the same output. Calculate i) gain without feedback ii) feedback factor β .
- 5) Draw the basic negative feedback amplifier which will smoothly input and output impedances to match microphone and loudspeaker respectively.

(7)

3/00 0.05 10%

- a. Draw a feed-back block diagram and show that
- b. Show that $\frac{AVF}{A} = \frac{1}{1+AB}$ and the feed-back factor is $\frac{1}{1+AB}$
- c. If the gain is 10, find the gain with feed-back.

d. If the output factor of an amplifier is 10, what must be its feed-back factor to achieve a closed loop gain of 8.